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THE IMPACT OF FINANCIAL EDUCATION FOR YOUTH IN GHANA

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ABSTRACT

We evaluate, using a randomized trial, two school-based financial literacy education programs in government-run primary and junior high schools in Ghana. One program integrated financial and social education, whereas the second program only offered financial education. Both programs included a voluntary after-school savings club that provided students with a locked money box. After nine months, both programs had significant impacts on savings behavior relative to the control group, mostly because children moved savings from home to school. We observed few other impacts. We do find that financial education, when not accompanied by social education, led children to work more compared to the control group, whereas no such effect is found for the integrated curriculum; however, the difference between the two treatment effects on child labor is not statistically significant.

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1 Introduction

Governments and donors often support policies to promote financial literacy with the aim of improving households' financial decisions. Financial literacy is defined as one's ability to understand financial concepts, plan one's finances, and understand financial services and products. While financial literacy is correlated with more prudent financial decisions and the use of formal savings and insurance products (Xu and Zia 2012), this correlation does not imply that teaching financial literacy will lead to more prudent financial behavior. Perhaps as a result of a presumed causal relationship, a multitude of financial literacy programs have emerged over the past several decades spanning a variety of content and delivery mechanisms.

Many financial literacy programs target youth. Even though children are under the financial umbrella of their parents, the hypothesis is simple: teaching financial literacy to children may more effectively shape long-term behaviors than teaching such skills later in life. If lessons taught during childhood persist during adulthood, investing in financial literacy for children may be a cost-effective way to achieve long-lasting impacts on financial decision making. There is, however, an often-discussed potential downside of introducing children to the world of finance too early: encouraging children to think more about money may lead them to prioritize income-generating activities at the expense of schooling (Varcoe et al. 2005). This concern leads some financial education programs for youth to also include social values and other such material to mitigate unintended negative consequences.

Despite the potential tradeoff, financial literacy programs for children are common. For example, the Banking on Our Future program in South Africa promotes financial literacy, entrepreneurship, and youth empowerment through school programs (Operation HOPE 2014). In Peru, the Financial Education Program for Secondary Students focuses on training teachers to disseminate knowledge of financial services to their students who subsequently transmit that knowledge to their families at home (OECD International Gateway for Financial Education 2007). In Somalia, financial literacy programs targeting youth rely on mass media, soap opera broadcasts, and mobile phones to teach children about saving and other aspects of finance (Xu and Zia 2012).

Although there is significant policy interest in youth financial education, little is known about its impact, particularly in developing countries, or about effective approaches for mitigating the potential consequence of reduced school attendance. We address this knowledge gap by testing the impact of two school-based financial literacy programs in Ghana. The first program followed a curriculum developed by Aflatoun.

Aflatoun is a large, international non-governmental organization (NGO) that has developed school-based curricula for financial literacy training and provides technical assistance to local partners, usually NGOs or ministries of education, to implement these curricula.² Its program has been implemented in over 100 countries to date. The Aflatoun program includes financial education, social education, and a school savings club. The social education component focuses on personal exploration and children's rights and responsibilities, while also highlighting the pitfalls of youth labor, such as forgoing school to work and the risk of dangerous working conditions.

We compare the impact of Aflatoun's program against a second program, the Honest Money Box (HMB), which was designed for this evaluation and is directly modeled after the financial components of Aflatoun's program, while omitting the social components. HMB thus focused strictly on improving financial skills and savings behavior. This treatment design allows us to evaluate the marginal benefits of the social component of the Aflatoun program when added to the financial literacy component.

We conducted the study during the 2010-2011 school year in 135 primary and junior high schools in southern Ghana. Schools were randomly assigned to receive either the full Aflatoun program (45 schools), the Honest Money Box program (45 schools), or control (45 schools). We measured a variety of outcomes, including financial decision-making, support for savings at home, labor, risk and time preferences, financial literacy, consumption, confidence, and academic performance.

Both programs had positive and significant impacts on savings behavior: children in both programs increased savings at school. However, we see no impact on the percentage of children who save nor on the total amount saved, suggesting that the programs led students to shift existing savings into school, or that children's self-reports of their savings volume are noisy and unreliable. We find no evidence for impacts on savings attitudes, home savings support, risk aversion, time preference, financial literacy, expenditures, confidence, or academic performance.

Critical to a key policy question, we find that the HMB program, but not the Aflatoun program, led youth to work more, although the difference between the two estimates is not statistically significant. School attendance did not change, which suggests a possible shift away from leisure or home production instead. We do not have direct evidence of a reduction in these alternative activities, however.

² See <http://aflatoun.org/>.

Our paper contributes to the limited body of rigorous evidence on the effectiveness of youth financial literacy training.³ Studies on the effects of financial literacy programs on primary and middle school children are especially scarce. Hinojosa et al. (2009) conduct a randomized evaluation of a financial literacy program in the United States for children in grades 4-10 and find positive impacts on mathematics scores and financial knowledge, although the analysis does not account for substantial attrition and non-compliance in the sample. Several non-experimental studies have found positive impacts of financial literacy training in primary and middle schools using comparisons of participants with non-participants, or using before-after comparisons of participants (Diem et al. n.d.; Harter and Harter 2007; Sherraden et al. 2011; Hagedorn, Schug, and Suiter 2012).

A somewhat larger literature evaluates the impact of financial literacy education at the secondary level. Bruhn et al. (2013) evaluate the impact of a financial education program in Brazilian public high schools. The three-semester program consisted of 72 case studies, each involving one to two hours of teaching. The authors find positive effects on financial proficiency (about 0.2 standard deviations), saving for purchases, and financial budgeting in data collected four and 16 months after the start of program implementation. Lührmann, Serra-Garcia, and Winter (2014) find that participation in financial education can increase time consistency among German high school students. Becchetti, Caiazza, and Coviello (2013) and Becchetti and Pisani (2012) evaluate a financial literacy program for high school students in Italy and find some evidence that the program increased financial literacy, although the analysis is complicated by large improvements in financial literacy in the control group and by pre-existing differences between treatment and control students. In non-experimental work in the US, Cole and Shastry (2010) use variation in state-mandated programs to identify the effects of financial literacy education in high school. They find no evidence that exposure to financial literacy education affects later savings. A number of other non-experimental studies have found mixed evidence on the effects of financial literacy training on high school students (Carlin and Robinson 2010; Mandell and Klein 2009; Varcoe et al. 2005; Walstad, Rebeck, and MacDonald 2010).

2 Program description and evaluation design

2.1 Program Description

The Aflatoun curriculum was developed by the international NGO and has been adapted and implemented around the world. The HMB curriculum was adapted from Aflatoun by Ask Mama Development

³ For a review of evaluations of financial literacy training for adults in developing countries, see Hastings, Madrian, and Skimmyhorn (2013); Miller et al. (2013).

Organization (AMDO) and Innovations for Poverty Actions (IPA) staff, and derived its name from the money box used to safeguard the savings deposits of club members. It contained the financial but not the social components of the Aflatoun treatment (see Table 1).

The HMB curriculum began with eight structured one-hour sessions conducted by teachers who acted as facilitators for school clubs set up as part of the program. The clubs met weekly for one hour after school. The content and objectives of the sessions are provided in Table 1. Children were first encouraged to participate in the club and, once recruited, were introduced to the importance of money, savings and spending, planning and budgeting, personal finances, and entrepreneurship.

The additional social components of the Aflatoun treatment included sessions on personal exploration and children's rights and responsibilities.⁴ Details on the additional sessions are listed in Table 2. For example, the Aflatoun curriculum taught children the rights described in the United Nations Convention on the Rights of the Child: "Children (under the age of 16 years) are entitled to be protected from social or economic exploitation and shall not be employed in or required to perform work that is likely to be hazardous or to interfere with their education or to be harmful to their health or physical, mental, spiritual, moral or social development." The curriculum also included several stories about children who were forced to work instead of attending school. These stories emphasized the difficult and dangerous working conditions experienced by children and encouraged them to see child labor as a violation of their basic rights. In part because the Aflatoun curriculum contained lessons dedicated these social topics, it was designed to take more time to cover than the HMB curriculum (around 24 hours in total).

After clubs had completed the Aflatoun or HMB curriculum, they continued to operate as savings clubs where children could deposit or withdraw their savings. Both programs provided the schools with a metal padlocked savings box which was used to safeguard children's deposits. Each deposit and withdrawal was recorded by the teacher or a student club officer in the club ledger book and in the member's passbook. The proper use of these tools was monitored by the implementing organizations throughout the study period. At baseline, before the intervention, none of the schools had after-school programs related to savings.

Both programs were implemented by the same Ghanaian organizations.⁵ The local organizations and international NGOs also coordinated with the Ghana Education Service, a government agency.

⁴ The curriculum was taught at different levels for primary and junior secondary students but covered the same set of core concepts. In program schools that contained both primary and junior secondary grades, children were typically divided into separate clubs by age.

⁵ The contracting partner was the Netherlands Development Organization who in turn partnered with Women and Development Project, the Ask Mama Development Organization, Berea Social Foundation, and Support for Community Mobilization Projects and Programs.

The interventions began in October 2010 and lasted through the close of the school year, in July 2011. In workshops on club curriculum and protocols, IPA and local organizations trained the teachers selected by their schools to lead an Aflatoun or HMB club. They also monitored program implementation throughout the study period by visiting schools and interviewing teachers and students about the progress and activities of the club. Timing of implementation varied across schools. Out of the 83 Aflatoun and HMB schools for which monitoring data are available, the majority established clubs in December 2010 and January 2011. By the end of February 2011, 72 schools (87%) had established a club.

2.2 Evaluation design

We exploited the intended phase-in of the Aflatoun program to employ an experimental design.⁶ From a list of 165 eligible schools⁷ located in the program districts provided by district officials and implementing partners, 135 were randomly selected to be included in the program. The sample includes primary (grades 1-6), junior high (grades 7 and 8), and “basic” (combined primary and junior secondary) schools in three districts: 36 in Nkwanta, 30 in Greater Accra East, and 69 in Sekondi Takoradi Metropolitan Area. Within each district, sample schools were sorted by average within-grade class size and then grouped into triplets. Within these triplets, schools were randomly assigned to the Aflatoun intervention, the HMB intervention, or a control group.⁸ Thus, there were 45 strata in the randomization.

Baseline data were collected in September 2010, and endline data nine months later in July 2011.⁹ We thus present short-term impacts of the Aflatoun and HMB programs.

We sampled an average of 40 students from each school in the study.¹⁰ Although children of all grades were eligible to participate in the after-school clubs, our surveys targeted children in grades 5 and 7 because these children would presumably have more access to money and familiarity with finances than their younger peers. Additionally, these students would remain in the same schools the following school year, and would

⁶ The intended phase-in did not materialize. As a result of budget issues, the program was not extended to control group schools.

⁷ Two exclusion criteria were applied: First, we excluded “shift” schools from the study sample. Unlike “straight day” schools, shift schools host two different groups of students in the morning and afternoon, making it difficult to implement an after school program. Second, when multiple schools were located within the same compound, we randomly selected only one of those schools to join the pool of potential study schools.

⁸ The randomized assignment was implemented correctly in all but two schools: one school assigned to the Aflatoun treatment implemented the HMB treatment, and one school assigned to the HMB treatment implemented the Aflatoun treatment. The analysis is based on the original randomized assignment.

⁹ Surveys are available online at <http://poverty-action.org/project/0465>.

¹⁰ In 118 schools, we surveyed exactly 40 students. Due to surveyor error or logistical constraints, we surveyed between 22 and 39 students in ten schools, and between 41 and 47 students in six schools.

be easier to locate if a follow-up occurred the next year.¹¹ In primary and junior high schools, 40 students were randomly selected from grades 5 and 7, respectively. In basic (combined) schools, 20 students were randomly selected from grade 5, and 20 were selected from grade 7. When schools contained fewer than the target number of students in a given grade, additional students were randomly selected from adjacent grades. The final sample contains 45% from grade 5, 46% from grade 7, and 9% from adjacent grades.¹² To deal with multiple hypothesis issues, we group the outcome indicators into 11 indices and discuss the components of each index in the results section. The 11 indices cover savings behavior, savings attitudes, home savings support, work, risk preference, time preference, financial literacy, expenditures on self, expenditures on temptation goods, confidence, and academic performance.

For each of the indices we follow the method employed by Kling, Liebman and Katz (2007). The summary index for child i over the set of N_d outcome variables in group d is defined as the mean of the z-scores of the non-missing outcome variables in that group. Missing outcome variables are imputed at the mean z-score of the non-missing variables for that individual. Each variable is scaled such that it contributes positively to the header or overall concept used for the index.

$$\tilde{y}_{id} = \frac{1}{N_d} \sum_{d=1}^{N_d} \frac{y_{id} - \bar{y}_d}{\sigma_d} \quad (2)$$

Where $\bar{y}_d$ and σ_d are the mean and standard deviations of variable y_{id} estimated from the control group schools. The resulting index $\tilde{y}_{id}$ is then normalized by subtracting the mean and dividing by the standard deviation from the control group. The final summary index thus provides an equal weight to each component variable and has a mean of zero and a standard deviation of one.

Table 3 presents summary statistics, including verification of orthogonality of treatment assignment with baseline values. Overall, there are few significant differences between the treatment groups at baseline. As shown in Column 7, two out of the 11 indices are not balanced at the 10 percent level across both the Aflatoun and HMB treatment groups (work index and temptation expenditures index). All impact specifications include controls for the stratification variables as well as the baseline value of the outcome measure, if it exists.

Attrition rates for the endline survey were low (1.4%) and uncorrelated with assignment to treatment.¹³ To test for differential attrition by treatment status along baseline characteristics, we regress completion of

¹¹ Students typically change schools after grade 6, hence our reason for excluding them from the survey sample.

¹² The main results are robust to restricting the sample to only 5th and 7th graders.

¹³ The attrition rate was 1.4% in the control schools, 1.4% in the Aflatoun schools, and 1.3% in the HMB schools.

endline survey on Aflatoun and HMB treatment dummies, the full set of baseline indices, and the indices interacted with each treatment dummy. The F-test that the treatment dummies and all interaction terms are jointly equal to zero has p-value = 0.55 (result not shown in table). We thus find no evidence that attriters have different baseline characteristics across treatment groups.

To obtain the impact estimates we employ a regression model

$$y_{ijk,endline} = \alpha_k + \beta_1(Aflatoun_{jk}) + \beta_2(HMB_{jk}) + \gamma y_{ijk,baseline} + \varepsilon_{ij} \quad (1)$$

where y_{ijk} denotes the outcome of student i in school j in strata k , α_k represents a dummy variable for each stratum, and $Aflatoun_{jk}$ and HMB_{jk} are dummies indicating the school's inclusion in either the Aflatoun or the HMB treatment. Standard errors are clustered at the school level, the unit of randomization. When outcome variables were not included in the baseline survey, $\gamma y_{ijk,baseline}$ is omitted from the specification. Where baseline values are missing for some but not all observations, we recode the missing baseline value to zero and include a dummy variable to indicate the missing value. When we do not have a full set of baseline values for components of an index, we construct the baseline index using only the components included in the baseline survey.

The impact estimates are intent-to-treat effects, and do not take into account whether the child participated in the savings club or not. An instrumental variable approach, to estimate the treatment on the treated, would require precise measures on participation in savings clubs in all of the schools in the study. We do not have such measures. Even if participation data were available, the instrumental-variables estimation would require assuming no impact on non-participants in treatment schools. Such an assumption would be difficult to substantiate because the theory of change of the program includes spillovers: untreated individuals influence the attitudes and thus behaviors of their fellow students (although we do not find a direct effect on savings attitudes). Furthermore, aside from technical obstacles to the treatment on the treated, the intent-to-treat estimate provides the more policy relevant estimate of the average impact of programs such as these.

Although we do not have data on club membership and attendance data for all schools in our sample, we were able to collect membership information for a non-random subset of ten Aflatoun schools and seven HMB schools.¹⁴ Below we present an analysis of the characteristics of children who took up the programs

¹⁴ The subsample of schools for which we have take-up data had similar patterns of implementation to those in the full sample. As in the full sample, the majority of schools in the take-up subsample established clubs in December 2010 and January 2011.

in these schools. We emphasize that because we do not have complete take-up information for all schools, these results should be taken as suggestive.

In the subgroup of schools for which we have data, 16 percent of the children surveyed had joined the clubs in the Aflatoun schools, and 30 percent of the children had joined the clubs in the Honest Money Box schools. Table 4 regresses an indicator for club membership on baseline values of our outcome indices as well as a set of seven demographic and academic variables. Column 1 restricts the sample to the Aflatoun schools for which we have data. The explanatory variables in this regression have little predictive power. Out of the 16 variables in the regression, the only statistically significant variables are school grade repetition and durables ownership (both positive, and statistically significant at the 10 and 5 percent level, respectively). Column 2 repeats the analysis for the HMB schools. In this case, students who save more at baseline are significantly more likely to be members of the HMB clubs, as well as students who are more financially literate and those who spend more. This suggests that interest in the HMB clubs could depend on prior experience with savings and money.

3 Results

Table 5 presents the impact of the programs on each of the 11 summary endline indices. Appendix Tables 1 through 11 show the results for the individual variables used to construct the indices.

3.1 Savings

The savings behavior index includes eight variables that measure the proportion of children who save, amounts saved, savings inside and outside of school, and regularity of savings. As shown in Table 5, we find positive impacts on the index for both programs, with HMB leading to an increase of 0.16 standard deviations (s.e. = 0.058), and Aflatoun producing a 0.12 standard deviation (s.e. = 0.053) increase. The difference between the Aflatoun and HMB program is not statistically significant ($p = 0.48$).

Appendix Table 1 shows the effects of the programs on each component of the savings behavior index. Both programs show positive effects on the proportion of children that save at school (9 percentage points, s.e. = 0.015, for HMB and 5 percentage points, s.e. = 0.015, for Aflatoun) and the amount of money children have saved at school (0.47 Ghana cedis¹⁵, s.e. = 0.14, for HMB, and 0.44 Ghana cedis, s.e. = 0.17, for Aflatoun; control mean = 0.165). For each program, the increase in the percentage of children who save is not significant (2.41 percentage points, se = 0.022 for Aflatoun, 3.15 percentage points, s.e. = 0.025 for

¹⁵ The exchange rate from Ghana cedis to USD was 1.5 at the time of the endline survey, in July 2011.

HMB), and neither is it significant when both treatments are pooled. We do not find any impact on total amount saved, but the 95% confidence interval on that variable is large (the upper bound of the treatment effect of Aflatoun is 25% of the control group mean). Nonetheless, a lack of an effect on the total savings suggests that the program caused students to move their savings to the school accounts. This is consistent with the fact that we do not find any impact on the expenditure variables.

The savings attitude index captures children's opinions on the importance of savings. The index is constructed from ten questions, nine of which are Likert-style questions where the respondent indicates level of agreement with a statement on a scale from one (strongly disagree) to four (strongly agree). Three statements relate to the student's general view of savings, four relate to whether the student believes s/he should save in addition to adults, and one question measures whether the student saves whenever possible. The final component of the index is the student's allocation to savings if s/he were hypothetically given five cedis. As shown in Table 5, we find a precise null pooled treatment effect of 0.031 standard deviation (s.e. = 0.039). Appendix Table 2 presents similar null results on each component.

The home savings support index reflects how the student's family perceives the student's savings, as well as access to savings at home. The five component variables measure whether the student talks to relatives about savings, how adults in the household view child savings, perceived safety of savings with family, and the number of household bank accounts. As shown in Table 5, we find a precise null pooled treatment effect of 0.012 standard deviations (s.e. = 0.04), and Appendix Table 3 presents similar null results on all but one component: we find a positive impact on the perception of students in the HMB group that their parents would be proud of them for saving, significant at the 10 percent level.

3.2 Labor Supply

Neither treatment encouraged children to seek paid work, but the Aflatoun program explicitly discouraged child work. For the Aflatoun program, we thus have competing forces: the emphasis the Aflatoun program put on planning for the future and child self-esteem may lead children to prioritize education over work, but the emphasis on savings and financial matters could result in children thinking proactively about work as a way, for instance, to accumulate savings. Because the HMB program did not include the social component, we hypothesize that the HMB program will increase work through the second mechanism.

The work index includes 11 variables measuring incidence of work, intensity of work, and earnings. As shown in Table 5, we find that the HMB program led to a 0.102 standard deviation (s.e. = 0.056) increase in this index. The estimate for the Aflatoun program is 0.038 standard deviations (s.e. = 0.05). However, the t-test comparing Aflatoun and HMB fails to reject equality (p-value = 0.26).

Appendix Table 5 disaggregates the effects on the different components of the work index. To put the results in context, it is important to first note that many children work. In the control group, 24 percent of children reported having worked for money in the past four months (February to May). The HMB program led to a 4.23 percentage point increase in the likelihood of engaging in any work (s.e. = 0.025) during that period, whereas we see no effect in the Aflatoun group (0.014, s.e. = 0.022, but the p-value for the test to reject equality of Aflatoun and HMB is 0.25). The same pattern is found when looking month by month. The increase for the HMB program was statistically significant in two out of the four months, whereas the change for Aflatoun was not significant in any month (p-values for difference across treatments are 0.14, 0.12, 0.07 and 0.14 for each of the four months). However, the increased work participation in the HMB group did not appear to lead to extra earnings in the thirty days prior to the survey (1.024 Ghana cedis, s.e. = 1.68).

The difference in reported labor between the Aflatoun and HMB treatments could arguably be driven by misreporting (i.e., an experimenter demand effect in which those in the Aflatoun treatment group, because of the treatment, underreport their child labor but do not actually change their labor supply). However, we posit this to be unlikely given that we do not observe any differential results on other outcomes that would also plausibly induce experimenter demand effects, if indeed the children perceived a benefit to misreporting. For example, we find no evidence of impacts on savings attitudes, even though both programs promoted a positive view of savings.

3.3 Risk and Time Preferences

We next examine two indices measuring risk and time preferences. The risk preference indicators employ standard hypothetical risk games, where the student is asked to choose between a certain outcome, and an uncertain outcome with a higher expected return. The time preference indicators are also based on a hypothetical game, where the student is asked to make trade-offs between income now and income in the future. Both indices also contain components that ask about risk and patience in hypothetical but more realistic situations.

Both treatments, through the promotion of entrepreneurship, may lead participants to feel more comfortable taking risks. On the other hand, the encouragement of long term planning and savings may encourage taking fewer risks. Thus the predicted impact on risk preferences is theoretically ambiguous. However, for time preferences, the prediction is less theoretically ambiguous: we expect the treatments to lead children to place greater value on future outcomes and thus display more patient time preferences.

Our risk preference index is constructed from three hypothetical choices between risky and safe bets, a self-reported scale of the child’s willingness to take risks, and the child’s hypothetical preference to start a high-risk, high-return business over a low-risk, low-return business. The impact estimates are shown in Table 5 and Appendix Table 5. We do not observe statistically significant changes in the risk preference index for either program (pooled results for the index is -0.07 standard deviations, s.e. = 0.049). However, in both Aflatoun and HMB schools, we observe statistically significant decreases in one component: children’s self-reported willingness to take risks. This question asked students, “Are you generally very prepared to take risks or do you try to avoid taking risks?” Students answered on a scale of 0 to 10, which we converted to a range of 0 to 1 for the analysis. Students in both the Aflatoun and HMB schools had responses 3 percentage points lower than control-group students (s.e. = 0.016 and s.e. = 0.014, respectively)

We measure time preference through two hypothetical inter-temporal choices and one question on whether the child would prefer to wait for a medicine that heals completely or receive a medication now that doesn’t heal completely. We find no statistically significant changes in time preferences from either of the treatments (Table 5 and Appendix Table 6; 0.032 standard deviations for the pooled treatment analysis, s.e. = 0.043).

3.4 Financial Literacy and Control of Spending

We now turn to measures of financial literacy. Financial literacy was measured through two hypothetical “shop games” in which the child was given a list of goods and prices and a certain amount of money, all of which had to be spent on the available goods. The child was then asked to report how much of each item s/he would buy. For each game, the index includes an indicator of whether the child correctly allocated the money (i.e., spent exactly the amount of money given), the absolute value of the difference between the child’s allocation and the correct allocation, and the number of seconds taken to respond. We also include an indicator of whether the student makes a spending plan each week. The results are shown in Table 5 and Appendix Table 7. The effects of the programs on the financial literacy index are small and not statistically significant (0.0052, s.e. = 0.049), and none of the seven individual components of the index show statistically significant effects.

Table 5 and Appendix Table 8 examine the student’s propensity to spend on temptation goods, based on three variables measuring actual and hypothetical spending on snacks and entertainment. We find no evidence for treatment effects on the index (Table 5; for pooled treatment, -0.027 standard deviations, s.e. = 0.042). Among the individual components of the index, the Aflatoun treatment reduced hypothetical spending on entertainment by 0.14 cedis (s.e. = 0.059), but there are no other statistically significant results.

We next examine control of personal spending by the child with an expenditure index, consisting of two questions on the amount the child spent on him/herself in the past seven days and the amount s/he expects to spend in the next seven days. We do not find a statistically significant impact on the expenditure index (Table 5; -0.04 standard deviations for the pooled treatment analysis, s.e. = 0.043) or on either question individually (Appendix Table 9).

3.5 Child Confidence

Table 5 and Appendix Table 10 display the program impacts on measures of confidence. We include five Likert questions that measure self-esteem and confidence at school. We find no evidence of impacts, though point estimates on the aggregate index are negative for both programs (-0.029 standard deviations for the pooled treatment analysis, s.e. = 0.038). Across all of the individual measures, the only measure that is significantly different in the treatment groups (10 percent level of significance) is an increased likelihood of agreeing with the statement “Teacher makes you feel you are not good enough” in Aflatoun schools. While this result could reflect a lower sense of confidence among the Aflatoun group, it should be interpreted tentatively, as no other indicator within the index shows statistically significant impacts.

3.6 Academic Performance

Finally, we examine program impacts on school attendance and achievement. Attendance was measured through self-reports of attendance over the past week. To measure aptitude, students were given ten-question tests in English and math. Separate tests were given to 5th- and 7th-graders, although the structure of the tests was similar. Test scores were normalized based on the baseline means and standard deviations in each grade. As shown in Table 5 and Appendix Table 11, we find no evidence of program effects on the combined academic performance index, or on either of the components individually (-0.04 standard deviations for the pooled treatment analysis, s.e. = 0.06).

4 Conclusion

We evaluate two programs in Ghana that aimed to increase financial literacy among youths: the Aflatoun program, which offered both financial and social education, and the HMB program, which offered only financial education. We find that both programs positively influenced savings behavior (which is explicitly facilitated through a locked savings box as part of the program), but we find fairly precise null results for impact on savings attitudes, home support for savings, risk and time preferences, spending patterns,

confidence, and academic performance. Savings attitudes and home support for savings are process changes, intended by program design to be a necessary step for behavior change.

The fact that we observe large behavior change but no underlying process changes is important. We posit two interpretations. First, simply put, it could be that changing attitudes and home environment was not a necessary step because a pro-savings attitude and environment already existed. What lacked was merely an infrastructure for the children to act on those attitudes and environmental factors, and the program provided that infrastructure. An alternative, pessimistic interpretation is that the behavior change was a mere artifact of the intervention, an attempt by the schools and children to follow along with a program (i.e. by substituting savings at home for savings at school), but, with no underlying change in values and attitudes, one that will dissipate in the long run.

A key test of this will be long-term results. If the optimistic interpretation is correct, one should see continued savings behavior in the long-run, as long as the infrastructure to save remains in place. If, on the other hand, the pessimistic interpretation is correct, behavior will revert and no long-term benefits will accrue. We also find important, although borderline statistically, results on the impact of the programs on child labor supply. One interpretation for the results on savings and work between the Aflatoun and HMB treatments is that the social curriculum in the Aflatoun program counteracted an increased interest in working brought about by the financial education curriculum. Thus, child labor may increase if social education is not included in a financial literacy program. We note, however, that the child labor we measure did not appear to displace schooling.

In our context, the costs of developing and implementing school-based financial education were modest. Excluding fixed curriculum development costs, which amounted to \$3,100, the marginal cost of implementing the HMB program was approximately \$5.53 per student. We expect that implementation costs of Aflatoun were similar, though we are unable to verify due to lack of precise data from the implementing organization. Because of the scarcity of evidence on financial literacy programs on this age group, we are unable to make explicit cost comparisons with other programs.

While our work provides a useful starting point for understanding the effects of youth financial education, more work is needed to both broaden the evidence base and understand the mechanisms behind the program effects. Again, this calls for long-term tracking of child financial education programs. The results we find, along with the modest cost of implementing school-based financial education, make these interventions worthy of continued study.

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Table 1: Honest Money Box Curriculum

Core Elements	Objectives
Form Club	Explain the function and operation of the money box club. State rules for club functioning. Identify leaders, elect President, Treasurer, and Secretary and assign roles and responsibilities.
What is money?	Explain money as a medium of exchange. Identify honest ways of making money.
Saving and Spending	Understand: The purpose of saving. How to save. Types of saving, including non-monetary resources. Responsible spending behavior.
The money box	Understand: Features of the money box, procedures for depositing and withdrawing. How to record transactions.
Planning and budgeting	Understand financial goals and develop their own financial goals. Create a budget plan.
Entrepreneurship	Understand: Business organization. Types of businesses. Skills necessary for running a business.

Table 2: Additional Elements of Aflatoun Curriculum

Core Elements	Objectives
Character and Motto	Orient children to the Aflatoun value framework, and enhance their creativity, problem-solving, and reasoning skills. Encourage children to learn more about Ghana and its unique cultural heritage. Facilitate an understanding among children that they can contribute to their environment, by teaching about the contributions made by different people and things.
Personal Understanding and Exploration	Enable children's positive self-image through self-awareness and appreciation, and highlight the different factors which contribute towards building self-image. Provide children an opportunity to assess themselves and then discuss the experience of being their own judge. Allow children to express their likes and dislikes in a non-threatening environment, and facilitate an understanding of the differences and similarities among people.
Rights and Responsibilities	Teach children a sense of responsibility for their actions towards everything and everyone in their environment, and an understanding that everything and everyone needs to be treated with respect. Orient children to their rights as described in the United Nations Convention on the Rights of the Child. Create awareness of the various marginalized groups who do not get their rights in Ghana and around the world, and develop a sense of responsibility towards those whose rights are violated. Sensitize children to the issues of working children and provide children an opportunity to interact with working children, thereby facilitating a process of dispelling myths and stereotypes. Sensitize children to issues related to gender and create awareness on the different forms of gender discrimination. Identify social projects and campaigns that could improve children's communities.

Table 3: Baseline Orthogonality

	Means and Standard Errors			p-value from t-test or F-test					Obs.
	Control group (1)	Aflatoun (2)	HMB (3)	(1) vs (2)		(2) vs (3)		F-test (7)	
				(1) (4)	(2) (5)	(1) (4)	(3) (6)		
Savings Behavior Index (higher = greater propensity to save)	0.000 (0.041)	-0.096 (0.036)	-0.083 (0.041)	0.055	0.131	0.848	0.130	5364	
Savings Attitudes Index (higher = more positive attitude towards savings)	0.000 (0.052)	0.075 (0.050)	0.004 (0.045)	0.203	0.931	0.239	0.375	5362	
Home Savings Support Index (higher = home environment is more conducive to)	0.000 (0.040)	-0.015 (0.039)	0.024 (0.035)	0.794	0.621	0.449	0.755	5364	
Work Index (higher = more likely to work, more hours, etc.)	0.000 (0.059)	-0.102 (0.037)	0.017 (0.050)	0.099	0.833	0.034	0.068	5364	
Risk Preference Index (higher = less risk averse)	0.000 (0.044)	0.037 (0.040)	0.072 (0.040)	0.514	0.218	0.500	0.469	5354	
Time Preference Index (higher = lower discount rate of the future)	0.000 (0.039)	0.013 (0.032)	0.008 (0.037)	0.843	0.876	0.952	0.975	5355	
Financial Literacy Index (higher = greater financial literacy)	0.000 (0.059)	0.053 (0.042)	0.016 (0.049)	0.449	0.827	0.588	0.726	5364	
Expenditures on Temptation Goods Index (higher = less propensity to spend on temptation goods)	0.000 (0.039)	-0.114 (0.032)	-0.047 (0.049)	0.018	0.405	0.242	0.057	5364	
Expenditures on Self Index (higher = higher expenditures on goods for self)	0.000 (0.059)	-0.001 (0.049)	0.054 (0.099)	0.913	0.581	0.597	0.854	5352	
Academic Index (higher = higher school attendance and performance)	0.000 (0.066)	0.016 (0.046)	-0.066 (0.070)	0.842	0.444	0.313	0.589	5364	
Completed Endline Survey (%)	0.986 (0.003)	0.986 (0.003)	0.987 (0.003)	0.935	0.964	0.904	0.993	5364	

Columns (4) - (7) present p-values for t-tests or F-tests of means using regressions of each row variable on Aflatoun and HMB treatment dummies. Regressions control for stratification dummies (region and standardized average class size).

Table 4: Characteristics affecting program takeup

	Aflatoun	HMB	Combined
	(1)	(2)	(3)
Savings Behavior Index	0.0121 (0.0282)	0.101*** (0.0197)	0.0444* (0.0234)
Home Savings Support Index	-0.0355 (0.0263)	-0.00563 (0.0298)	-0.0213 (0.0198)
Work Index	-0.0234 (0.0172)	0.0103 (0.0423)	-0.000426 (0.0234)
Risk Preference Index	0.0160 (0.0196)	0.00696 (0.0325)	0.0110 (0.0195)
Time Preference Index	-0.00581 (0.0186)	-0.0165 (0.0398)	-0.0102 (0.0215)
Financial Literacy Index	0.0258 (0.0170)	0.0933* (0.0419)	0.0554** (0.0229)
Expenditures on Temptation Goods Index	0.0267 (0.0352)	0.0141 (0.0296)	0.0331 (0.0236)
Expenditures on Self Index	0.0304 (0.0310)	0.0181** (0.00619)	0.0163*** (0.00468)
Academic Performance Index	0.00402 (0.0143)	0.0102 (0.0413)	-0.000192 (0.0185)
Female	0.0991 (0.0638)	0.134 (0.0772)	0.120** (0.0476)
Age	0.00619 (0.0179)	-0.0341 (0.0213)	-0.0223 (0.0167)
Ever repeated grade	0.0773* (0.0383)	0.0535 (0.0431)	0.0801** (0.0318)
Index of durable good ownership	0.0238** (0.00927)	-0.0311 (0.0365)	0.000412 (0.0218)
Household (HH) size	-0.00866 (0.00962)	-0.0140 (0.0207)	-0.0125 (0.00778)
Number of earners in HH	0.0266 (0.0304)	0.00778 (0.0378)	0.0239 (0.0245)
Household wages per week / 100	-0.00873 (0.00682)	0.00103 (0.0107)	-0.00182 (0.00807)
Mean of dependent variable	0.162	0.297	0.217
R-squared	0.0642	0.131	0.0680
Number of observations	328	241	569
Number of Schools	10	7	17

Takeup is defined as attendance at one or more Aflatoun or HMB club meetings, as indicated by the club roster sheet or attendance logs. Row variables are measured at baseline. Each column presents the results of an OLS regression of takeup on the row variables in the Aflatoun and/or HMB schools for which club rosters or attendance logs were collected. Index of durable good ownership is constructed using First Principal Component Analysis. Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 5: Treatment Effects on Indices of Key Outcome Variables

Outcome Variable	Aflatoun (1)	Honest Money Box (2)	p-value, Afla = HMB (3)	Pooled Effect (4)	Obs (5)
Savings Behavior Index (higher = greater propensity to save)	0.119** (0.0531)	0.164*** (0.0583)	0.479	0.141*** (0.0457)	5291
Savings Attitudes Index (higher = more positive attitude towards savings)	0.0134 (0.0433)	0.0490 (0.0479)	0.468	0.0312 (0.0386)	5291
Home Savings Support Index (higher = home environment is more conducive to saving)	-0.0267 (0.0496)	0.0516 (0.0494)	0.134	0.0123 (0.0423)	5291
Work Index (higher = more likely to work, more hours, etc.)	0.0377 (0.0495)	0.102* (0.0564)	0.257	0.0699 (0.0449)	5291
Risk Preference Index (higher = less risk averse)	-0.0645 (0.0544)	-0.0763 (0.0541)	0.804	-0.0704 (0.0487)	5291
Time Preference Index (higher = lower discount rate of the future)	0.0325 (0.0488)	0.0308 (0.0518)	0.975	0.0317 (0.0427)	5291
Financial Literacy Index (higher = greater financial literacy)	0.0154 (0.0554)	-0.00508 (0.0566)	0.714	0.00519 (0.0486)	5291
Expenditures on Temptation Goods Index (higher = less propensity to spend on temptation goods)	-0.0330 (0.0478)	-0.0216 (0.0442)	0.766	-0.0273 (0.0419)	5291
Expenditures on Self Index (higher = higher expenditures on goods for self)	-0.0156 (0.0505)	-0.0645 (0.0458)	0.287	-0.0400 (0.0425)	5291
Confidence Index (higher = more confident)	-0.0468 (0.0448)	-0.0108 (0.0445)	0.456	-0.0288 (0.0377)	5291
Academic Performance Index (higher = higher school attendance and test score)	-0.0328 (0.0641)	-0.0467 (0.0644)	0.798	-0.0398 (0.0583)	5291

Columns (1) and (2) present individual regressions of each index on Aflatoun and HMB treatment dummies. Column (4) presents individual regressions of each index on dummies for either HMB or Aflatoun treatment. Regressions control for stratification dummies (region and standardized average class size) and baseline values for the index if available. Indices are aggregated ignoring missing values in the individual variables. Standard errors clustered at the school level, in parentheses. Money amounts reported in Ghana cedis. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 1: Savings Behavior

Dependent Variables:	Saving Behavior Index	Has money saved right now	Total money saved right now (GHC)	Has money saved right now at school	Total money saved at school right now (GHC)	Has money saved outside school right now	Total money saved outside school right now (GHC)	Regularly saves money during the week	Amount saved last week (GHC)
	(1)	(2)	(3)	(5)	(6)	(7)	(8)	(9)	(10)
Panel A: Individual Treatment Effects									
Aflatoun	0.119** (0.0531)	0.0241 (0.0219)	-0.287 (1.299)	0.0520*** (0.0147)	0.440*** (0.166)	-0.00718 (0.0222)	-0.575 (1.381)	0.0196 (0.0200)	-0.274 (0.218)
HMB	0.164*** (0.0583)	0.0315 (0.0248)	-1.091 (1.264)	0.0913*** (0.0151)	0.474*** (0.141)	-0.00897 (0.0225)	-0.580 (1.355)	0.00116 (0.0205)	-0.265 (0.224)
Control mean	0.000	0.555	9.121	0.0276	0.165	0.528	9.033	0.363	1.233
Control std. deviation	1.000	0.497	38.55	0.164	2.541	0.499	36.48	0.481	6.538
Baseline mean of outcome variable	-0.059	0.452	4.622					0.410	0.917
Observations	5291	5291	5291	5291	5291	5291	5291	5291	5291
R-squared	0.0529	0.0381	0.0504	0.0215	0.00807	0.0101	0.00590	0.0745	0.0151
p-value for test of Aflatoun = HMB	0.479	0.769	0.500	0.0464	0.858	0.938	0.997	0.378	0.937
Panel B: Pooled Treatment Effect									
Aflatoun or HMB	0.141*** (0.0457)	0.0278 (0.0198)	-0.688 (1.136)	0.0716*** (0.0114)	0.457*** (0.121)	-0.00807 (0.0191)	-0.577 (1.214)	0.0104 (0.0174)	-0.269 (0.214)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Index is aggregated ignoring missing values in the individual variables. Savings amounts (Columns 3, 6, 8, 10) are self-reported and in Ghana cedis. Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 2: Savings Attitudes

Dependent Variables:	Saving Attitude Index	Think that saving is good	They are happy if they save	Think that spending now is better than saving for the future †	Save every time they get money	Think that saving is for adults only †	Think that saving is for parents only †	Don't think they need to save because parents buy them what they need †	Think that they don't need to save if they're living at home †	Proportion allocated to saving in hypothetical spending exercise (11)
	(1)	(2)	(3)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Panel A: Individual Treatment Effects										
Aflatoun	0.0134 (0.0433)	-0.00657 (0.0274)	-0.00346 (0.0232)	-0.0162 (0.0269)	0.0423 (0.0276)	0.00565 (0.0266)	0.00441 (0.0257)	-0.0207 (0.0253)	-0.00816 (0.0258)	-0.00880 (0.0173)
HMB	0.0490 (0.0479)	-0.00243 (0.0238)	0.0163 (0.0251)	-0.0201 (0.0289)	0.00969 (0.0276)	-0.0127 (0.0259)	-0.0291 (0.0232)	-0.0507 (0.0309)	-0.0380 (0.0294)	-0.0215 (0.0177)
Control mean	8.70e-19	2.353	2.094	1.040	1.823	0.944	1.006	1.190	1.060	0.255
Control std. deviation	1.000	0.601	0.570	0.663	0.664	0.605	0.556	0.688	0.626	0.395
Baseline mean of outcome variable	0.027						0.724			0.269
Observations	5291	5287	5274	5285	5288	5284	5291	5290	5286	5281
R-squared	0.0273	0.00174	0.0157	0.00944	0.0319	0.00768	0.0117	0.0148	0.00724	0.0371
p-value for test of Aflatoun = HMB	0.468	0.868	0.391	0.889	0.218	0.536	0.205	0.322	0.315	0.429
Panel B: Pooled Treatment Effect										
Aflatoun or HMB	0.0312 (0.0386)	-0.00450 (0.0225)	0.00641 (0.0213)	-0.0181 (0.0240)	0.0260 (0.0243)	-0.00350 (0.0217)	-0.0123 (0.0208)	-0.0357 (0.0239)	-0.0230 (0.0234)	-0.0151 (0.0156)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Outcome variables in Columns (2) through (10) takes on integer values ranging from 0 (strongly disagree) to 3 (strongly agree). † indicates that the variable enters the index negatively. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Baseline index is aggregated ignoring missing values in the individual variables. Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 3: Home Savings Support

Dependent Variables:	Home Savings Support Index	Have talked to parents or relatives about the importance of savings in last 7 days	Someone in household would be angry if they found out student was saving for self †	Parents would be proud of them for saving	Perceived safety of saving with family (1 being least safe, 5 most)	Number of household bank accounts
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Individual Treatment Effects						
Aflatoun	-0.0267 (0.0496)	-0.0215 (0.0144)	0.0139 (0.0141)	0.0111 (0.0271)	0.00801 (0.0747)	0.00698 (0.0443)
HMB	0.0516 (0.0494)	0.0159 (0.0166)	-0.00354 (0.0129)	0.0491* (0.0256)	0.0533 (0.0739)	-0.0175 (0.0482)
Control mean	0.000	0.138	0.122	2.064	2.700	0.851
Control std. deviation	1.000	0.345	0.328	0.616	1.610	0.901
Baseline mean of outcome variable	0.005		0.177		1.988	0.761
Observations	5291	5287	5231	5263	5121	5291
R-squared	0.0529	0.00251	0.0369	0.00572	0.0172	0.228
p-value for test of Aflatoun = HMB	0.134	0.0198	0.235	0.174	0.485	0.616
Panel B: Pooled Treatment Effect						
Aflatoun or HMB	0.0123 (0.0423)	-0.00286 (0.0135)	0.00520 (0.0114)	0.0301 (0.0225)	0.0307 (0.0669)	-0.00523 (0.0394)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Outcome variables in Columns (3) and (4) take on integer values ranging from 0 (strongly disagree) to 3 (strongly agree). † indicates that the variable enters the index negatively. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Baseline index is aggregated ignoring missing values in the individual variables. Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 4: Work

Dependent Variables:	Work Index	Worked in past 4 months to earn money	Days worked in past 30 days	Amount of money earned		Worked in Feb	Worked in Mar	Worked in Apr	Worked in May	Worked inside household	Worked outside household	Worked "a lot" during school term
				30 days	past 30 days, winsorized at 95%							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Panel A: Individual Treatment Effects												
Aflatoun	0.0377 (0.0495)	0.0137 (0.0215)	0.247 (0.260)	2.226 (1.484)	0.350 (0.471)	0.000190 (0.0144)	0.00278 (0.0146)	0.00295 (0.0155)	0.00267 (0.0205)	-0.0145* (0.00851)	0.0234 (0.0197)	0.00510 (0.0107)
HMB	0.102* (0.0564)	0.0423* (0.0247)	0.675* (0.354)	1.024 (1.681)	0.337 (0.515)	0.0229 (0.0147)	0.0279* (0.0154)	0.0345* (0.0178)	0.0361 (0.0228)	0.0107 (0.0105)	0.0355 (0.0221)	0.0164 (0.0119)
Control mean	0.000	0.237	2.221	6.918	3.864	0.0884	0.0976	0.129	0.190	0.0603	0.188	0.0620
Control std. deviation	1.000	0.425	6.101	29.56	9.974	0.284	0.297	0.335	0.392	0.238	0.391	0.241
Baseline mean of outcome variable	-0.027	0.309	3.818	12.23	7.104					0.0814	0.244	0.100
Observations	5291	5291	5291	5291	5291	5291	5291	5291	5291	5291	5291	5291
R-squared	0.0375	0.0504	0.0275	0.0126	0.0352	0.0135	0.00986	0.00281	0.00906	0.0129	0.0481	0.00931
p-value for test of Aflatoun = HMB	0.257	0.249	0.226	0.528	0.981	0.138	0.119	0.0742	0.139	0.00777	0.581	0.325
Panel B: Pooled Treatment Effect												
Aflatoun or HMB	0.0699 (0.0449)	0.0280 (0.0196)	0.461* (0.257)	1.626 (1.271)	0.343 (0.423)	0.0115 (0.0125)	0.0153 (0.0128)	0.0187 (0.0143)	0.0193 (0.0186)	-0.00189 (0.00845)	0.0294 (0.0179)	0.0108 (0.00978)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Outcome in Column (2) includes tasks or chores, either inside or outside the household, to earn money. Outcome variable in Column (5) censors the top 5% of observations of earnings variable. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Baseline index is aggregated ignoring missing values in the individual variables. Standard errors clustered at the school level, in parentheses. Money amounts reported in Ghana cedis. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 5: Risk Preference

Dependent Variables:		Would choose to play a game getting 6 cedis win and 0 cedis lose rather than a game getting 3 cedis win or lose		Would choose to play a game getting 6 cedis win and 0 cedis lose rather than a game getting 2 cedis win or lose		Would choose to play a game getting 6 cedis win and 0 cedis lose rather than a game getting 1 cedi win or lose		Self-reported willingness to take risks (0 low risk aversion to 1 high risk aversion)		Would start a high risk-high return rather than low risk-low return business	
		(1)	(2)	(3)	(4)	(5)	(6)				
Panel A: Individual Treatment Effects											
Aflatoun	-0.0645 (0.0544)	-0.0225 (0.0229)	-0.0265 (0.0229)	0.00616 (0.0242)	-0.0342** (0.0155)	0.00660 (0.0196)					
HMB	-0.0763 (0.0541)	-0.0218 (0.0210)	-0.0305 (0.0219)	-0.0154 (0.0228)	-0.0250* (0.0140)	0.000984 (0.0185)					
Control mean	0.000	0.346	0.429	0.535	0.515	0.202					
Control std. deviation	1.000	0.476	0.495	0.499	0.308	0.401					
Baseline mean of outcome variable	0.036					0.209					
Observations	5291	5287	5287	5290	5288	5285					
R-squared	0.0217	0.0253	0.0262	0.00843	0.0313	0.0154					
p-value for test of Aflatoun = HMB	0.804	0.973	0.833	0.343	0.556	0.763					
Panel B: Pooled Treatment Effect											
Aflatoun or HMB	-0.0704 (0.0487)	-0.0221 (0.0195)	-0.0285 (0.0204)	-0.00460 (0.0206)	-0.0296** (0.0125)	0.00380 (0.0167)					

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Baseline index is aggregated ignoring missing values in the individual variables. Standard errors clustered at the school level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Appendix Table 6: Time Preference

Dependent Variables:	Time Preference Index	Prefer 9 cedis in one week to 6 cedis now	Prefer 9 cedis in five weeks to 6 cedis in four weeks	Rather wait for medicine that heals completely than take one now that doesn't completely heal
	(1)	(2)	(3)	(4)
Panel A: Individual Treatment Effects				
Aflatoun	0.0325 (0.0488)	-0.0115 (0.0197)	0.00956 (0.0184)	0.0309 (0.0227)
HMB	0.0308 (0.0518)	0.0109 (0.0186)	-0.00250 (0.0180)	0.0212 (0.0232)
Control mean	0.000	0.737	0.820	0.667
Control std. deviation	1.000	0.441	0.384	0.471
Baseline mean of outcome variable	0.008			0.620
Observations	5291	5291	5290	5286
R-squared	0.00633	0.00254	0.00203	0.0113
p-value for test of Aflatoun = HMB	0.975	0.293	0.533	0.683
Panel B: Pooled Treatment Effect				
Aflatoun or HMB	0.0317 (0.0427)	-0.000335 (0.0160)	0.00354 (0.0154)	0.0261 (0.0197)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Baseline index is aggregated ignoring missing values in the individual variables. Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 7: Financial Literacy

Dependent Variables:	Difference between student's allocation and correct allocation in Shop Game 1 †				Difference between student's allocation and correct allocation in Shop Game 2 †				Do you make a plan for how to spend your money during the week?
	Financial Literacy Index	Allocation in Shop Game 1 correct	Seconds taken for Shop Game 1 †	Allocation in Shop Game 2 correct	Seconds taken for Shop Game 2 †	Allocation in Shop Game 2 correct	Seconds taken for Shop Game 2 †		
Panel A: Individual Treatment Effects									
Aflatoun	0.0154 (0.0554)	-0.0160 (0.0247)	0.0209 (0.0257)	-0.0713 (2.620)	0.0151 (0.0219)	0.00311 (0.0163)	0.899 (2.141)	0.0143 (0.0273)	
HMB	-0.00508 (0.0566)	-0.0160 (0.0274)	0.00257 (0.0253)	-1.039 (2.651)	0.00144 (0.0182)	-0.0137 (0.0167)	0.221 (2.106)	-0.0101 (0.0247)	
Control mean	0.000	0.248	0.444	44.05	0.129	0.843	39.49	0.654	
Control std. deviation	1.000	0.739	0.497	42.16	0.603	0.364	35.25	0.476	
Baseline mean of outcome variable	0.025	0.297	0.382	55.38	0.154	0.813	47.79	0.672	
Observations	5291	5291	5291	5291	5291	5291	5290	5282	
R-squared	0.0520	0.00542	0.00699	0.0725	0.000845	0.00323	0.0493	0.0303	
p-value for test of Aflatoun = HMB	0.714	0.999	0.462	0.692	0.534	0.353	0.736	0.343	
Panel B: Pooled Treatment Effect									
Pooled treatment effect	0.00519 (0.0486)	-0.0160 (0.0231)	0.0117 (0.0223)	-0.554 (2.336)	0.00830 (0.0169)	-0.00526 (0.0139)	0.561 (1.872)	0.00212 (0.0227)	

Two games were conducted as part of the survey, testing the ability of students to allocate money in hypothetical shopping scenarios. They were given a certain amount of money and a goods/price list then asked to allocate their money to purchase the goods. They were evaluated on whether they completely allocated the money, the amount of money left over, and how long they took. Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. † indicates that the variable enters the index negatively. In Columns (2) & (5), because students were asked to allocate all of the money, the greater the difference between a student's allocation and the correct allocation, the worse her performance on the financial literacy test. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 8: Expenditures on Temptation Goods

Dependent Variables:	Temptation Goods Index	Amount spent on snacks in the last 7 days	Amount spent on non-food goods and entertainment in the last 7 days	Amount would spend on fun if given 5 cedis
	(1)	(2)	(3)	(4)
Panel A: Individual Treatment Effects				
Aflatoun	-0.0330 (0.0478)	0.0361 (0.0551)	0.0241 (0.153)	-0.142** (0.0592)
HMB	-0.0216 (0.0442)	0.00340 (0.0614)	-0.128 (0.118)	-0.0116 (0.0499)
Control mean	0.000	0.586	0.719	0.666
Control std. deviation	1.000	1.261	3.487	1.416
Baseline mean of outcome variable	-0.051	0.644	0.560	0.359
Observations	5291	5291	5291	5291
R-squared	0.0523	0.0291	0.0130	0.0371
p-value for test of Aflatoun = HMB	0.766	0.565	0.255	0.0292
Panel B: Pooled Treatment Effect				
Aflatoun or HMB	-0.0273 (0.0419)	0.0197 (0.0510)	-0.0516 (0.119)	-0.0769 (0.0466)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Standard errors clustered at the school level, in parentheses. Money amounts reported in Ghana cedis. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 9: Expenditures on Self

Dependent Variables:	Expenditure Index	Amount spent on self in the last 7 days	Amount expects to spend in the next 7 days
	(1)	(2)	(3)
Panel A: Individual Treatment Effects			
Aflatoun	-0.0156 (0.0505)	0.0541 (0.307)	-0.281 (0.359)
HMB	-0.0645 (0.0458)	-0.193 (0.269)	-0.528 (0.336)
Control mean	0.000	5.249	5.964
Control std. deviation	1.000	5.700	8.288
Baseline mean of outcome variable	0.020	5.154	5.983
Observations	5291	5291	5286
R-squared	0.154	0.142	0.0935
p-value for test of Aflatoun = HMB	0.287	0.386	0.446
Panel B: Pooled Treatment Effect			
Aflatoun or HMB	-0.0400 (0.0425)	-0.0689 (0.252)	-0.404 (0.308)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Regressions control for stratification dummies (region and enrollment per stream) and baseline values of the dependent variable if available. Spending on self can include, for instance, money spent on food, clothes and school supplies. Standard errors clustered at the school level, in parentheses. Money amounts reported in Ghana cedis. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 10: Confidence

Dependent Variables:	Confidence Index	Confident in taking exams at school	Has a low opinion of self †	Often feels upset at school †	Teacher makes them feel they are not good enough †	Often gets discouraged at school †
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Individual Treatment Effects						
Aflatoun	-0.0468 (0.0448)	-0.0169 (0.0303)	0.0300 (0.0294)	-0.00649 (0.0246)	0.0448* (0.0263)	0.000437 (0.0250)
HMB	-0.0108 (0.0445)	-0.0219 (0.0289)	0.00883 (0.0311)	0.00212 (0.0219)	0.00398 (0.0227)	-0.0180 (0.0231)
Control mean	0.000	2.047	1.066	1.160	1.055	1.070
Control std. deviation	1.000	0.611	0.626	0.630	0.580	0.603
Observations	5291	5285	5281	5287	5281	5286
R-squared	0.00917	0.00731	0.0103	0.00250	0.00186	0.00381
p-value for test of Aflatoun = HMB	0.456	0.865	0.473	0.719	0.130	0.446
Panel B: Pooled Treatment Effect						
Aflatoun or HMB	-0.0288 (0.0377)	-0.0194 (0.0257)	0.0195 (0.0265)	-0.00220 (0.0200)	0.0244 (0.0207)	-0.00876 (0.0209)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. Individual outcome variables take on integer values ranging from 1 (strongly disagree) to 4 (strongly agree). † indicates that the variable enters the index negatively. Regressions control for stratification dummies (region and enrollment per stream). Standard errors clustered at the school level, in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Appendix Table 11: Academic Performance

Dependent Variables:	Academic Performance Index (1)	Days of school attended, last week (2)	Standardized aptitude test score (3)
Panel A: Individual Treatment Effects			
Aflatoun	-0.0328 (0.0641)	-0.0375 (0.0683)	-0.0291 (0.0651)
HMB	-0.0467 (0.0644)	-0.0970 (0.0653)	0.00527 (0.0663)
Control mean	0.000	4.493	0.0159
Control std. deviation	1	1.223	1.032
Baseline mean of outcome variable	-0.011	4.612	0.00
Observations	5291	4720	5291
R-squared	0.0476	0.0163	0.0781
p-value for test of Aflatoun = HMB	0.798	0.370	0.546
Panel B: Pooled Treatment Effect			
Aflatoun or HMB	-0.0398 (0.0583)	-0.0674 (0.0582)	-0.0120 (0.0593)

Each column in Panel A presents the results of an OLS regression of the outcome variable on Aflatoun and HMB treatment dummies. Each column in Panel B presents the results of an OLS regression of the outcome variable on a dummy for either HMB or Aflatoun treatment. The outcome variable in Column (3) takes the value of the student's standardized aptitude test score for either the primary or junior high school version of the aptitude test. The score distribution for each aptitude test was standardized within the relevant test-taking population, and these two sets of standardized scores were then combined to form one composite variable. Regressions control for stratification dummies (region and enrollment per stream). Standard errors clustered at the school level, in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$